School of Information
144 West 14th Street, 6th floor
New York, NY 10011–7301

INFO-664 Programming Cultural Heritage

Course Information

Section: 02
Semester: Fall 2026
Meeting Dates & Times: T 3:00 PM - 5:50 PM
Location: Manhattan - W14, 606 Lecture
Credits: 3
Prerequisites or other restrictions: None

Instructor Contact Information

Filipa Calado, PhD (she/her)
Office Location: Pratt Manhattan Campus, room 602
Office Hours: Mondays, 5:00pm - 6:00pm; Wednesdays 1:30-2:30pm (in person); and by appointment (on zoom)
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Website: https://filipacalado.com

Course Description

Technical knowledge and proficiency is an increasingly valued and useful skill in a cultural heritage information professional's skillset. This course will introduce and provide a platform for the building up of these abilities. Over the course of the semester the student will become familiar with programming, working with data in a programmatic manner and applying these skills to build a demonstrable project. This course will focus on teaching the basics that will enable an information professional to be successful in today's digital world.

Course Goals

- To introduce students to the technical solutions and abilities useful to an information professional.
- To foster and develop technical skills necessary for working with data in a cultural heritage context.
- To promote creative thinking and problem solving towards a targeted research question

through the application of gained skills and knowledge.

Student Learning Objectives

- Students will become familiar with basic programming in python.
- Students will develop the skills needed to work with data programmatically and at a large scale.
- Students will build additional ancillary technical skills that aids and supports the work of a cultural heritage information professional.

Course Format

This course will be conducted as part lecture and hands-on workshop with an emphasis towards working on a semester-long project. The class time will be divided between learning new skills through each class lecture and applying that knowledge in hands-on assignments.

Course Materials

Bringing a personal laptop to class will be necessary for participating in the lectures and workshops. Other materials will be provided electronically. Changes to the schedule will be communicated in advance, in class and via email.

Class Communication

The quickest avenue for contacting the instructor is via email, at **fcalado@pratt.edu.**

Because the email address that will be utilized to contact you is your Pratt email, please take the time to have your Pratt email forwarded to your primary email address in order to make sure that you receive all the information that is sent.

Assignments

Weekly Challenges

Weekly Challenges are meant to extend the skills learned in the class session on a new problem. Each student will submit their solution via Canvas. Emphasis is placed on attempting the challenge not necessarily finding a working or correct solution.

Final Project

Students will work on a final project for the class. These projects will align to the student's area of interest. More information on the final project and requirements can be found on the curriculum website (the course notebooks): https://gofilipa.github.io/664

Project pitches due before class on Week 11, and final projects are due on the last day of class.

Take a look at previous projects: https://gofilipa.github.io/664-projects

syllabus: https://bit.ly/664-fall26 | notebooks: https://bit.ly/info-664

Participation:

Each student is expected to participate in class discussions and workshops as well as complete readings and class exercises.

Course grades are determined as follows:

40% Weekly Challenge Assignments

10% Participation, Exercises and Workshops

10% Final Project Pitch

40% Final Project

Grading Policy

Superior work:	A 4.0 (96-100); A- 3.7 (90-95)
Very good work:	B+ 3.3 (87-89); B 3.0 (83-86); B- (80-82)
Marginally satisfactory:	C+ 2.3 (77-79); C 2.0 (70-76)
Failed:	F 0.0 (0-69)

Portfolio

Work completed for this course may be included in your portfolio. For more information on each program's portfolio requirements, please visit the program's respective webpage:

MS Library & Information Science: Portfolio - http://bit.ly/prattmslisportfolio

MS Information Experience Design: Portfolio - http://bit.ly/prattmsixdportfolio

MS Data Analytics and Visualization: Portfolio - http://bit.ly/prattmsdavportfolio2

MS Museums and Digital Culture: Portfolio - http://bit.ly/prattmsmdcportfolio2

You're final project would be a good possibility to include in your portfolio

For more information on Pratt Institute's Attendance Policy, please visit http://bit.ly/prattattendance.

syllabus: https://bit.ly/664-fall26 | notebooks: https://bit.ly/info-664

Tentative Course Schedule

Week 1 August 25	Introduction to the class Intro to the command line & to Python, day 1
Week 2 September 1 Python	Intro to Python day 2 **Reading due**: Benji Edwares, "Will the future of software development run on vibes?" https://arstechnica.com/ai/2025/03/is-vibe-coding-with-ai-gnarly-or-reckless-maybe-some-of-both/ **Homework Due:** Python 3+ installation on your computer through Anaconda
Week 3 September 8 Python	Python day 3 **Reading due:** "Declare Your AIndependence: Block AI Bots, Scrapers and Crawlers with a Single Click." *Cloudflare Blog*, 3 July 2024, https://blog.cloudflare.com/declaring-your-aindependence-block-ai-bots-scrapers-and-crawlers-with-a-single-click/. **Challenge 3 due:** Copy a few paragraphs of text from a web page that relates to your interests. Then use the methods we learned in class to do the following: first, transform the string into a list of individual words; then, add, remove, and sort items in the list. Save your python notebook file (.ipynb file) and submit it on canvas. **Due**: lists challenge
Week 4 September 15 Gathering: Web Scraping I	Introduction to web scraping with the BeautifulSoup4 library **Reading due:** two short articles, "The Quiet Way Gender Ideology Reaches Children" and "Confessions of a Male-Centered Woman" from *The Heritage Foundation* topic page on "Gender." **Challenge 4 due**: With a dataset from the NYC Open Data website, open it in Python using the CSV module. Then, use some loops and filtering logic to extract specific rows of data (that have certain strings or integers, for example). Run two or three different loops, checking for different conditions each time. Save your .ipynb file and submit it on canvas.
Week 5 September 22 Gathering:	Web scraping part two Working with HTML. **Challenge 5 due**: Find a web page that enables you to scrape the html (i.e. some web

Web Scraping II	pages have "blockers" installed: to make sure you can scrape it correctly, check if soup.title matches the web page's title). Using bs4, extract some content from that page, searching for the title element and another element (like paragraph or headings). Loop through an element using find_all() to extract some text you are interested in. At the bottom of your notebook, include a few sentences explaining why you are interested in this data, what you'd like to know more about it or do with it. You can write this reflection as a comment or in markdown (toggle "markdown" mode for the cell from the toolbar at the top of your notebook). Submit your notebook on canvas and be prepared to discuss your interest in this data in class.
Week 6 September 29 Gathering: APIs I	Introduction to APIs: the MET API; introduction to JSON **Challenge 6 due**: Using the website from the previous challenge, extract some data from that page and save that data to another file, like a csv or txt file. *If you run into errors and cannot complete the task, spend some time researching the error; then, write a paragraph at the bottom (you can write it as a code comment) about your challenges and what you learned from researching the error. Submit your notebook on Canvas.*
Week 7 October 6 Gathering: APIs II	APIs continued **Challenge 7 due**: Use the "search" path of the MET API (see Search path docs here) to do a search with at least two different parameters in the API call. Then, use one of the "objectIDs" from the results to write a new API call (that employs the "object path") to grab and display the details of a specific object. Submit your notebook to canvas.
Week 8: October 13	Indigenous Peoples' Day - No class
Week 9 October 20 Catch up	Catch up day: share APIs; discuss web errors **Challenge 9 due**: Find an API that relates to your interest (maybe checking the list of free APIs), and figure out how to use it to request some data. Write up a short (1-2 paragraphs) report on your API and how you used it. Come to class ready to briefly share your work with the API.
Week 10 October 27 Cleaning and manipulating data	Intro to OpenRefine (TBD – likely by Bryan Winston at PUL) **Challenge 11 due:** gather some data from either an API or web scraping that is related to what you want to work on for your final project. The data can be a web page, csv file, or text file --- whichever one is most appropriate for your project. Submit a python notebook with code that you used to get the data, which also shows an extract or sample of the data (like a slice from a string, list, or dictionary).

Week 11 November 3 Text generation	**Data gathering pitch** Text analysis with NLTK **Challenge 11 due**: write up a brief (1 - 2 paragraphs) pitch for how you will be gathering your data for the final project. Include specific sources and python libraries used. Share any concerns that you have about gathering this data. Finally, speculate on what you might do with it via text analysis, data analysis (which we will cover in depth in the next 2 weeks), or another kind of analysis (visual, narrative, etc).
Week 12 November 10 Analyzing and visualizing data	Data analysis and visualization with pandas **Challenge 12 due**: This challenge asks you to experiment with some data analysis and reflect. Using your dataset from last week, do something to your data, some kind of cleaning or manipulation of the data. Then, *write up a 1-2 paragraph reflection* on what you did and how it moves your project forward. You might identify questions for further data gathering or analysis. Submit your write-up on canvas.
Week 13 November 17 Publishing	**Project pitch day** Introduction to Github **Reading due**: Check out former class projects at https://gofilipa.github.io/664-projects/ **Project Pitch due**: write up a brief (1 - 2 paragraphs) project pitch. Identify at least one source where you will be gathering data, and how you plan to do so (web scraping, APIs, both?). Finally, speculate on what you'd like to do with that data once you get it, including any research questions or areas of potential analysis. Submit your write-up on canvas, and be ready to briefly share your project pitch with the class.
Week 14 November 24 CLASS IS ONLINE	Static sites with Github pages Workshop projects
Week 15 December 1	**READMEs Due**: Project READMEs: 1-2 pages, including all the info from "Complete Your README" in the online curriculum. Workshop projects
Week 16 December 8	Project presentations **Github links to project due at 11:59 pm**

Policies

The following abbreviated set of policies is especially relevant to this class. Full details on policies and procedures can be found on the Pratt website or in the Office of the Vice President for Student Affairs, Main Building, Lower Level.

Community Standards

All Pratt students, faculty, and staff members are expected to value and uphold the community standards essential to the pursuit of academic excellence and social responsibility. These include expectations for social conduct, academic integrity, non-discrimination, and other policies described in the link above, and apply to all Pratt-sponsored activities, on or off campus.

Academic Integrity

Academic integrity at Pratt means using your own and original ideas in creating academic work. It also means that if you use the ideas or influence of others in your work, you must acknowledge them.

At Pratt,

- We do our own work,
- We are creative, and
- We give credit where it is due.

When students submit any work for academic credit, they make an implicit claim that the work is wholly their own, completed without the assistance of any unauthorized person. These works include, but are not limited to exams, quizzes, presentations, papers, projects, studio work, and other assignments and assessments. In addition, no student shall prevent another student from making their work. Students may study, collaborate, and work together on assignments at the discretion of the instructor.

Examples of infractions include but are not limited to:
The following examples are drawn from the Academic Integrity Code and should be curated or supplemented based on assignments for your course.

1. Plagiarism, defined as using the exact language or a close paraphrase of someone else's ideas without citation.
2. Violations of fair use, including the unauthorized and uncited use of another's artworks, images, designs, etc.
3. The supplying or receiving of completed work including papers, projects, outlines, artworks, designs, prototypes, models, or research for submission by any person other than the author.
4. The unauthorized submission of the same or essentially the same piece of work for credit in two different classes.
5. The unauthorized supplying or receiving of information about the form or content of an examination.
6. The supplying or receiving of partial or complete answers, or suggestions for answers; or the supplying or receiving of assistance in interpretation of questions on any

examination from any source not explicitly authorized. (This includes copying or reading of another student's work or consultation of notes or other sources during an examination.)
7. The use of generative artificial intelligence (AI) to produce or to improve work, whether visual or textual, except when called for by an assignment or instructor and acknowledged transparently as one tool among others in the creative process.

The Academic Integrity Standing Committee (AISC) is charged with educating faculty, staff, and students about academic integrity practices. Whenever possible, we strive to resolve alleged infractions at the most local level possible, such as between student and professor, or within a department or school. When necessary, members of this committee will form an Academic Integrity Hearing Board to hear cases regarding cheating, plagiarism, and other infractions described below; these infractions can be grounds for citation, sanction, or dismissal. Detailed procedures are explained in the full version of the Academic Integrity Policy.

Attendance Policy

Consistent attendance is essential for the completion of any course or program. Attending class does not earn students any specific portion of their grade, but is the pre-condition for passing the course, while missing class may seriously harm a student's grade. Pratt Institute respects students' requirements to observe days of cultural significance, including religious holy days, and recognizes that some students might need to miss class to do so. In this, or other similar, circumstance, students are responsible for consulting with faculty ahead of time about how and when they can make up work they will miss. Reasonable accommodations for students with documented disabilities will continue to be provided, as appropriate. For full details, see Pratt's Attendance Policy.

The expectations for this class are as follows:

In graduate school, attendance to every class is crucial. You may miss up two classes (unexcused) with no penalty. Any more than that, your grade may be affected. In that case, please reach out to me.

Academic Support

Multiple academic support resources are available to students across campus:

- For assistance with time management and/or studio, subject, and software tutoring, contact the Student Success Center at success@pratt.edu.
- For assistance with writing assignments, contact the Writing and Tutorial Center at wtc@pratt.edu. The Pratt Libraries can also help with research and citations.
- Academic advisors are also a great resource; students can find their advisor's contact information or schedule an appointment through Starfish.

Accessibility

The Learning/Access Center (L/AC) coordinates access for students with disabilities. Students who identify as having any type of disability are entitled and encouraged to enroll with the L/AC in order to determine and implement reasonable accommodations. Contact the Learning/Access Center at lac@pratt.edu or 718.802.3123 for information or to schedule an appointment.

Title IX and Sexual Misconduct Policy

Pratt Institute is committed to fostering an environment that is safe, secure and free from sex discrimination and sexual harassment, sexual violence, dating and domestic violence, and stalking among all forms of sexual misconduct. The Institute takes prompt and appropriate action to address misconduct, end a hostile environment if one has been created, and prevent the recurrence of a hostile environment. To submit a concern, please use the Title IX and Sexual Misconduct Disclosure Form. For full details, see the Title IX and Sexual Misconduct Policy.

If you inform me of an issue of sex discrimination or sexual misconduct, I will keep the information as private as I can. However, as a faculty member, I am a mandatory reporter of sexual misconduct and required to bring it to the attention of the Institute's Title IX Coordinator, who can also be contacted at titleix@pratt.edu. You can also speak to someone confidentially by contacting our confidential resources in Health Services or the Counseling Center.

Nondiscrimination and Anti-Harassment Policy

Pratt is committed to fostering a welcoming, safe, non-discriminatory, and harassment-free educational, living, and working environment for its community. To submit a concern, please use the Discrimination and Harassment Disclosure Form, which also allows anonymous reports. For full details, see the Nondiscrimination and Anti-Harassment Policy.

Wellbeing

Pratt is dedicated to creating a culture where the entire community can flourish and thrive. Taking time to care for yourself and seeking appropriate support is important to achieving your academic and professional goals. Several resources are available through Starfish, including our Student Advocate and Care Coordinator, who can also be reached at 718.399.4546 or studentadvocate@pratt.edu. If you or anyone you know experiences overwhelming academic stress, persistent difficult feelings, or challenging life events, the Counseling Center can be reached 24/7 by calling 718.687.5356. To schedule a consultation, please call or email therapy@pratt.edu

Starfish

Starfish provides students with contact information for advisors, instructors, department chairs, and support services across the campus. Faculty can also use Starfish to inform students of their progress in class and connect them with resources. Students are contacted by support offices whenever a flag is raised.

Public Safety & Emergency Contacts

The Department of Public Safety provides 24-hour-a-day protection to the campus, including an emergency response guide. Contact Public Safety at security@pratt.edu or 718.636.3540. The Pratt Emergency Alert System is used to send urgent messages to registered mobile devices and emails (faculty, staff, and students have the option to opt-out, which must be renewed each year). You can update your emergency contact information in the Safety section of OnePratt.